

Unit 1.3 What is a function?

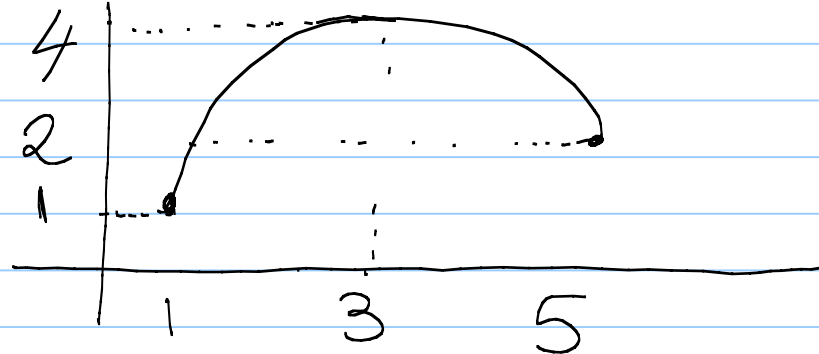
Note Title

2013/02/13

Stewart Section 1.1 p10-23

A function f is a rule that assigns to each element x in a set D exactly one element, called $f(x)$ in set E .

Example



Domain: $x \in [1, 5]$

Range: $y \in [1, 4]$

Find $f(1) = 1$

$f(3) = 4$

$f(5) = 2$

Finding Domain and Range of a function

We use formula of the function to find domain and the graph of the function to find the range of some functions.

- The domain and range are sets and the answer must be a set! ✓
Use interval notation or set builder notation.

Example Find domain of the function.

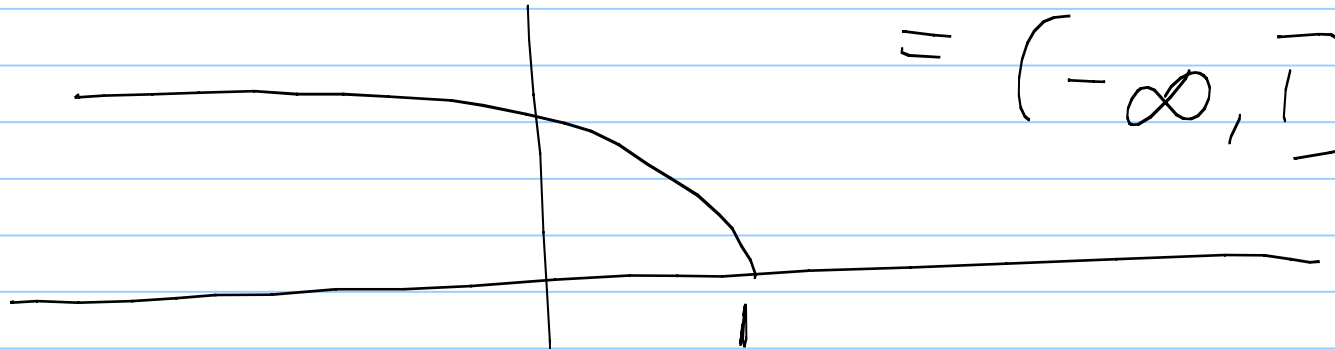
Sketch the function and find the range of the function.

① $f(x) = \sqrt{1-x}$

Domain: $1 - x \geq 0$

$$\Rightarrow 1 \geq x \Rightarrow \{x \mid x \leq 1\} = (-\infty, 1]$$

Sketch:



Range: $\{y \mid y \geq 0\} = [0, \infty)$

$$f: (-\infty, 1] \rightarrow [0, \infty)$$

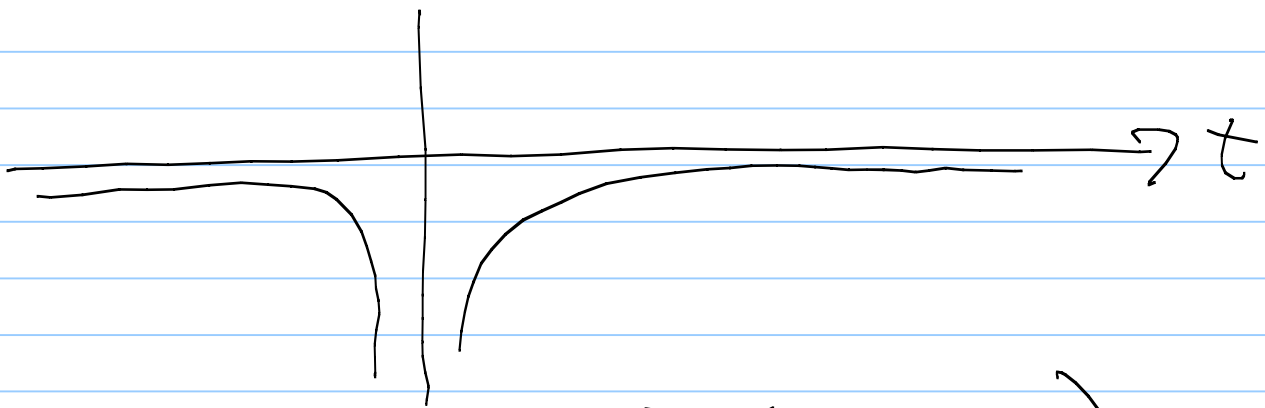
$$\textcircled{2} \quad f(t) = -\frac{1}{t^2}$$

Domain: $t \neq 0$

$$\Rightarrow \{t \mid t \in \mathbb{R} \setminus \{0\}\}$$

$$= (-\infty, 0) \cup (0, \infty)$$

Sketch:

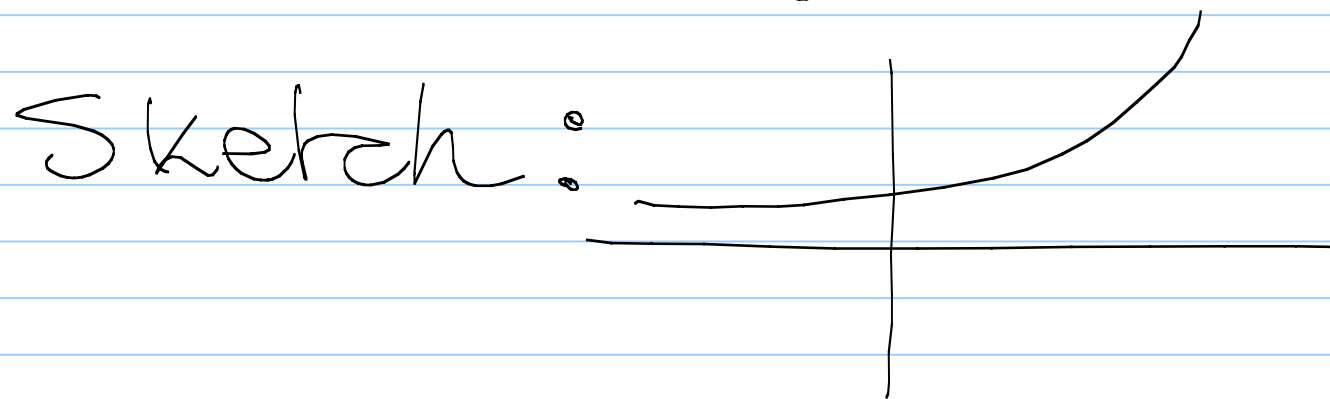


Range:

$$y \in (0, \infty) \cup (-\infty, 0)$$

③ $y = 3^u$

Domain : $\{u \mid u \in \mathbb{R}\}$



Range : $y \in (0, \infty)$

Example Find Domain

$$\textcircled{1} \frac{\sqrt{1-x}}{x^2-4}$$

$$\Rightarrow 1-x \geq 0 \quad \text{and} \quad x^2-4 \neq 0$$

$$\Rightarrow 1 \geq x \quad \text{and} \quad x \neq \pm 2$$

$$\Rightarrow x \in (-\infty, -2) \cup (-2, 1]$$

$$\{x \mid x \leq 1, x \neq \pm 2\}$$

$$\textcircled{2} \frac{\sqrt{x}}{\sqrt{3-2x}}$$

$$\Rightarrow x > 0 \quad \text{and} \quad 3-2x > 0$$

$$\Rightarrow x > 0 \quad \text{and} \quad x < 3/2$$

$$\Rightarrow x \in [0, 3/2)$$

$$\{x \mid 0 \leq x < 3/2\}$$

$$\textcircled{3} \quad \sqrt{\sqrt{x-2}-4}$$

$$\Rightarrow x-2 \geq 0 \text{ and } \sqrt{x-2}-4 \geq 0$$

$$\Rightarrow x \geq 2 \text{ and } \sqrt{x-2} \geq 4$$

$$\Rightarrow \begin{array}{l} x-2 \geq 16 \\ x \geq 18 \end{array}$$

$$\Rightarrow x \in [18, \infty) \cup \{x \mid x \geq 18\}$$

$$\textcircled{4} \quad f(x) = \frac{2x}{\sqrt{(2+x)(1-x)}}$$

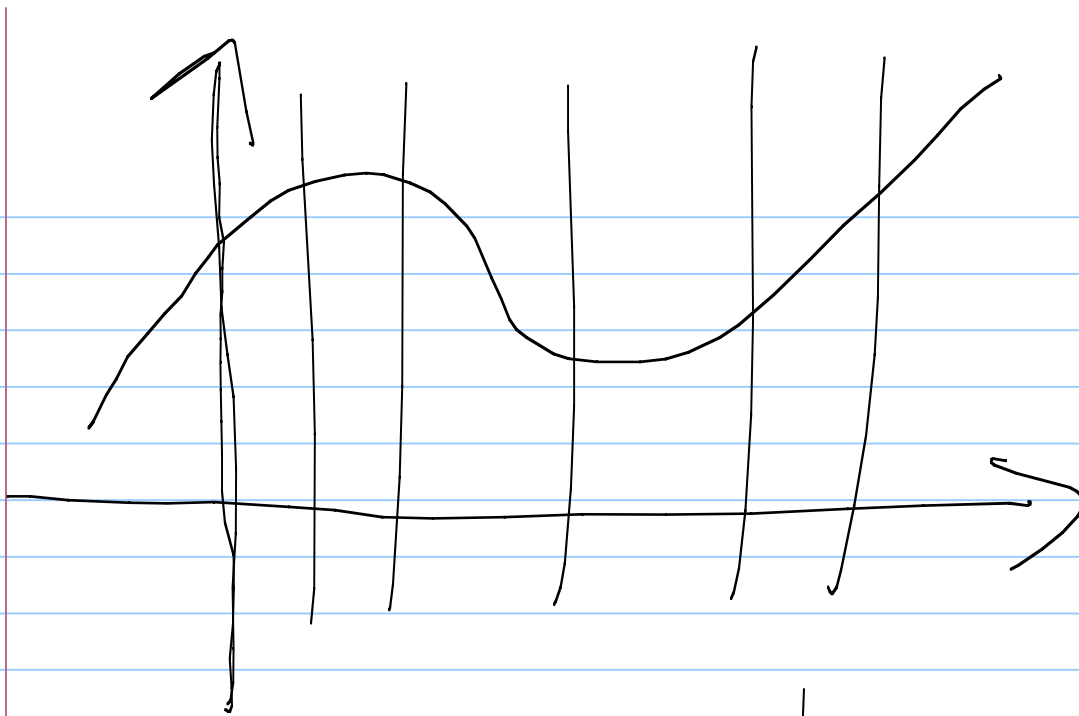
$$\Rightarrow 2+x > 0 \quad \text{and} \quad (1-x) > 0$$

$$\Rightarrow x > -2 \quad \text{and} \quad 1 > x$$

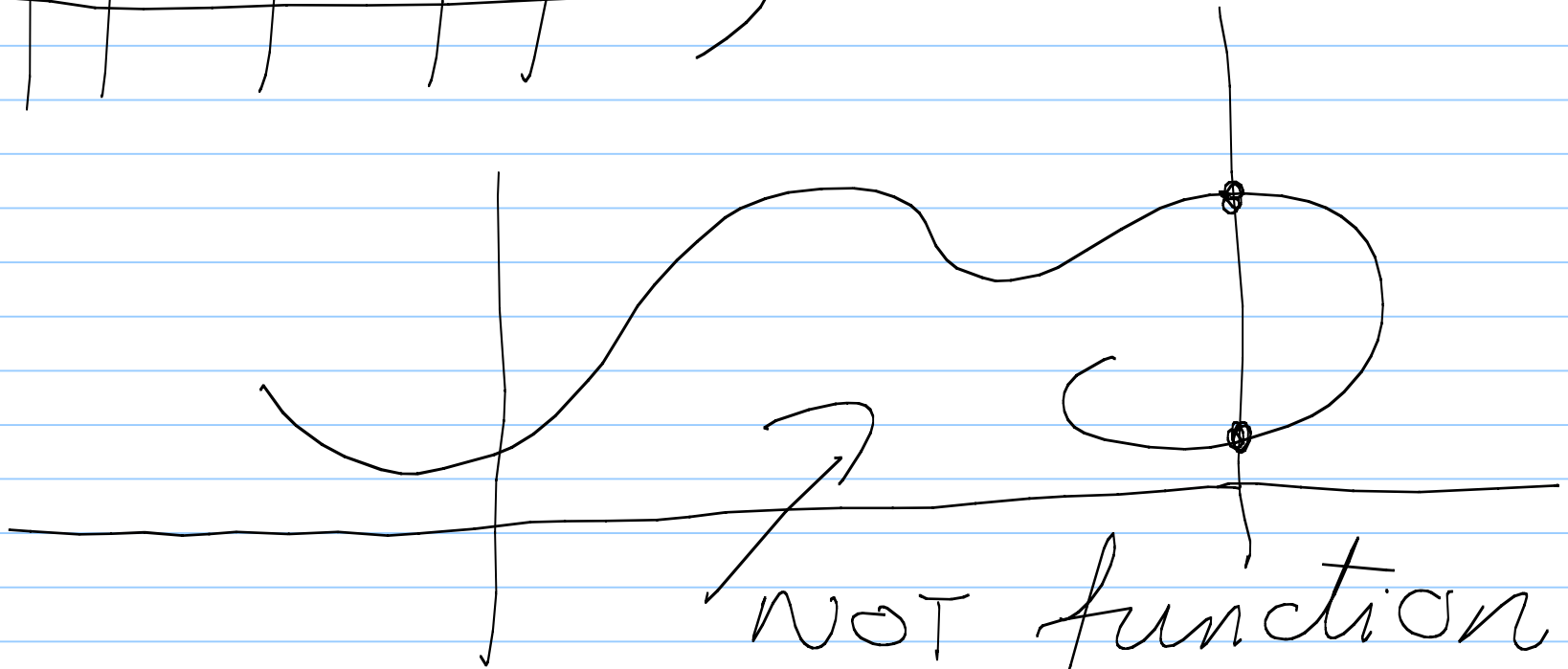
$$\Rightarrow \{x \mid -2 < x < 1\}$$

The Vertical line test p15

A curve in the xy -plane is the graph of a function of x if and only if no vertical line intersects the curve more than once



← function



↗ NOT function!

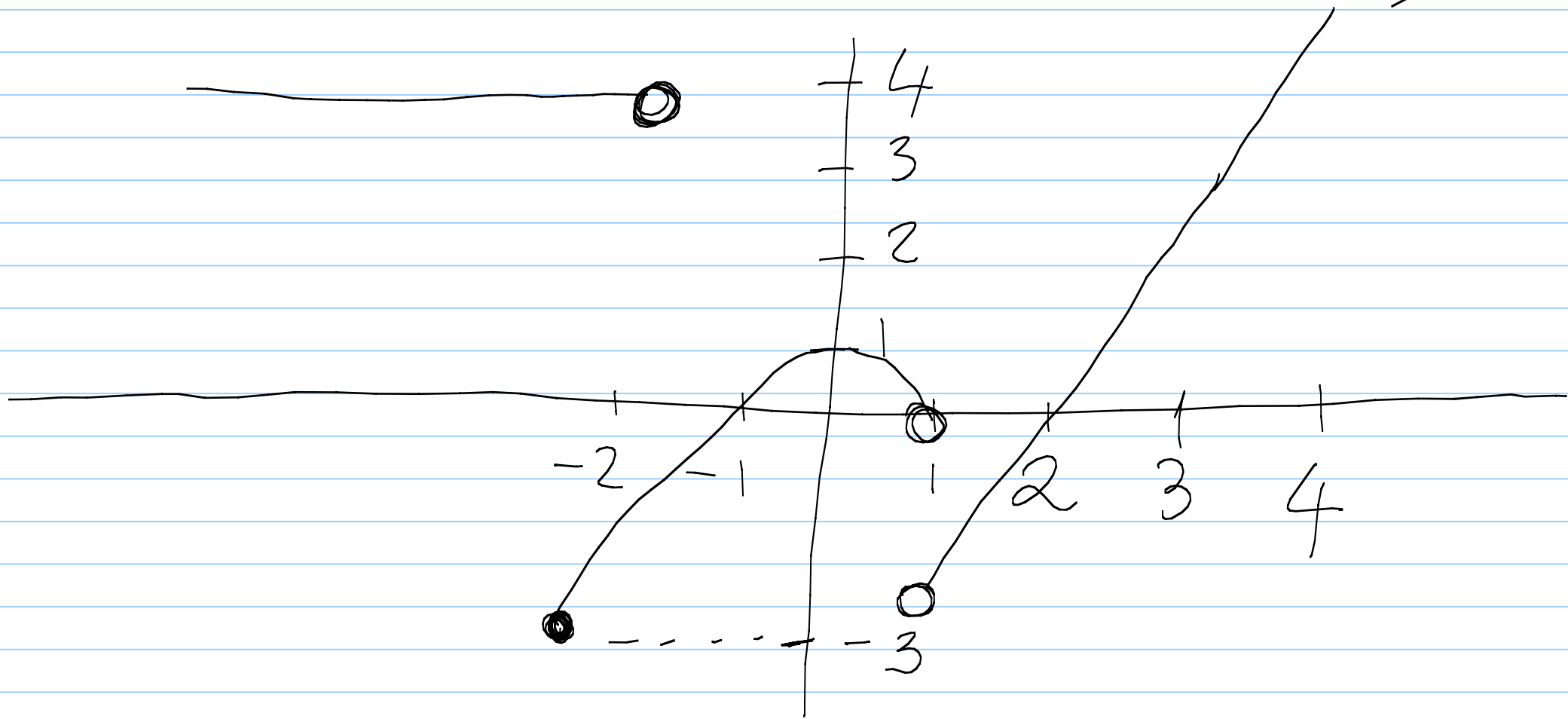


Piece-wise defined functions

Example Find the domain of the function
sketch the function and find range of
the functions

$$f(x) = \begin{cases} 4 & \text{if } x < -2 \\ -x^2 + 1 & \text{if } -2 \leq x < 1 \\ x - 2 & \text{if } x > 1 \end{cases}$$

Domain: $x \in (-\infty, 1) \cup (1, \infty)$



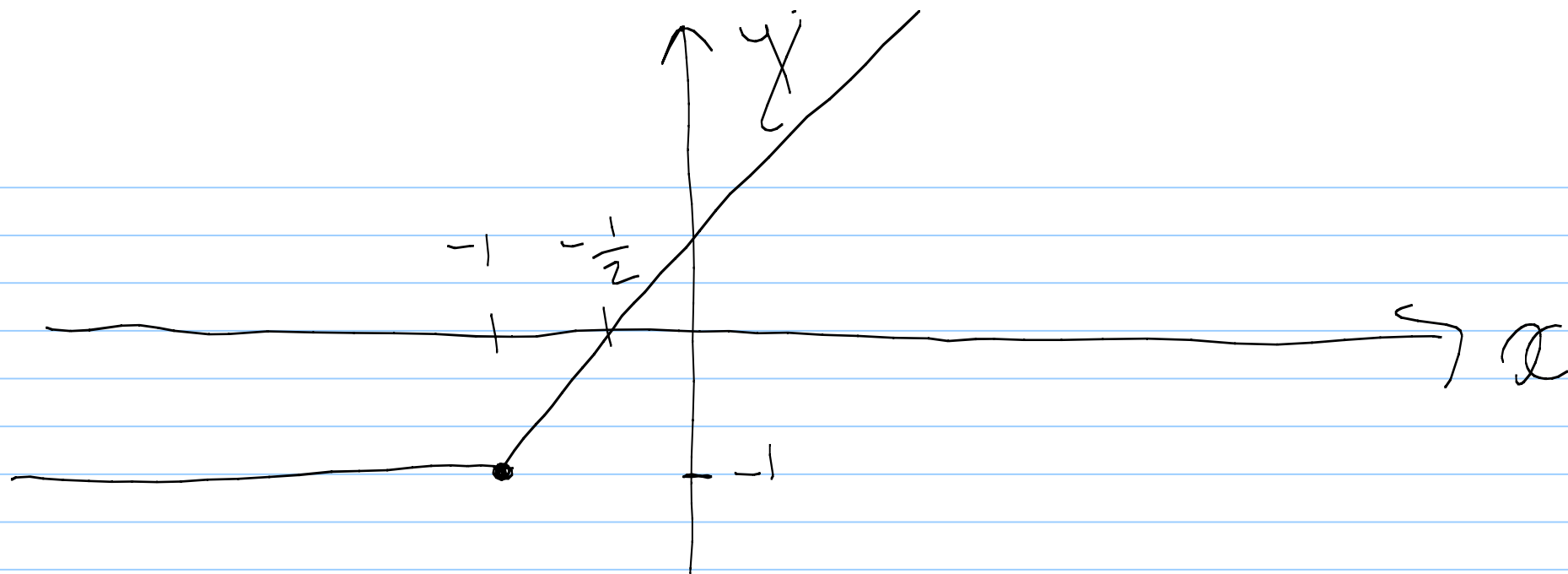
$$\text{Range: } \{y \mid y \geq -3\}$$
$$= [-3, \infty)$$

$$\textcircled{2} \quad f(x) = x + |x+1|$$

$$f(x) = x + |x+1|$$

$$= \begin{cases} x + x + 1 & x + 1 \geq 0 \\ x - x - 1 & x + 1 < 0 \end{cases}$$

$$= \begin{cases} 2x + 1 & x \geq -1 \\ -1 & x < -1 \end{cases}$$

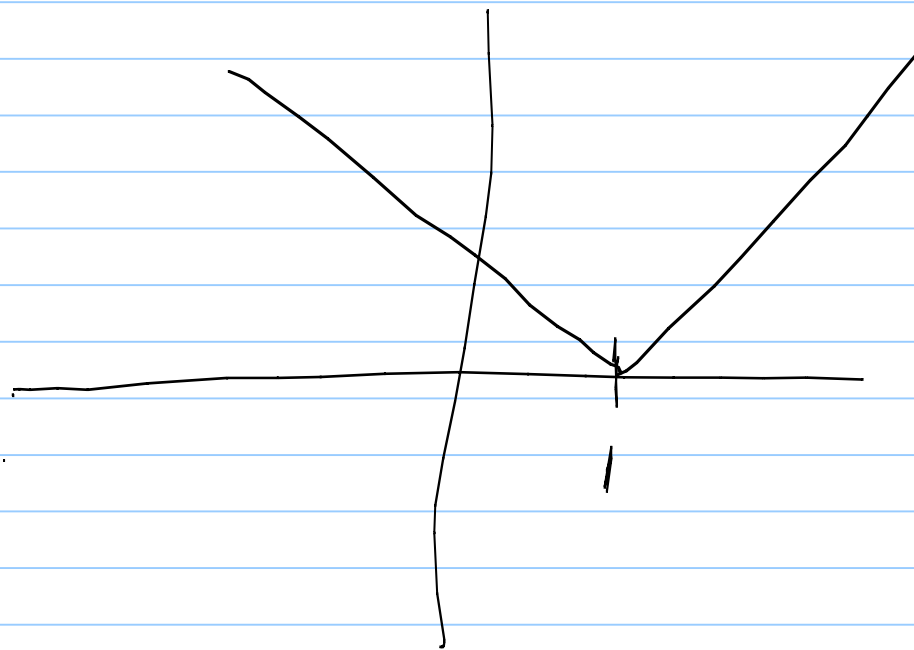
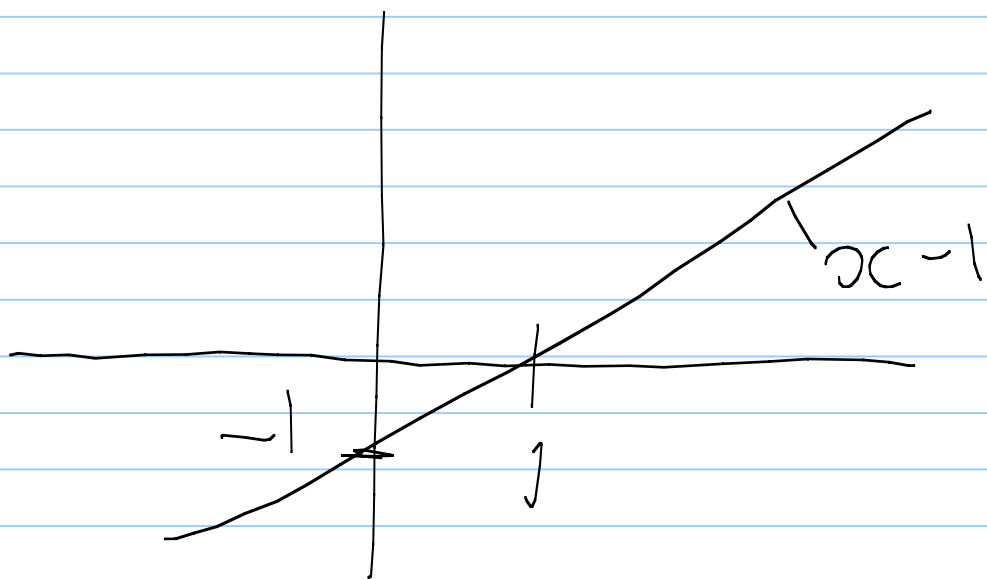


Domain : $\{x \mid x \in \mathbb{R}\} = (-\infty, \infty)$

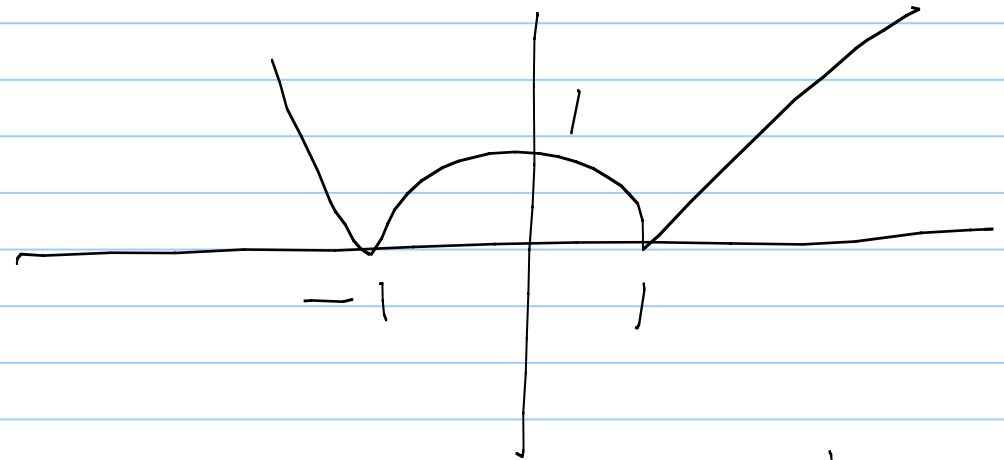
Range : $y \in [-1, \infty) = \{y \mid y \geq -1\}$

Example Sketch $f(x)$ and $|f(x)|$ without using the definition of the absolute value

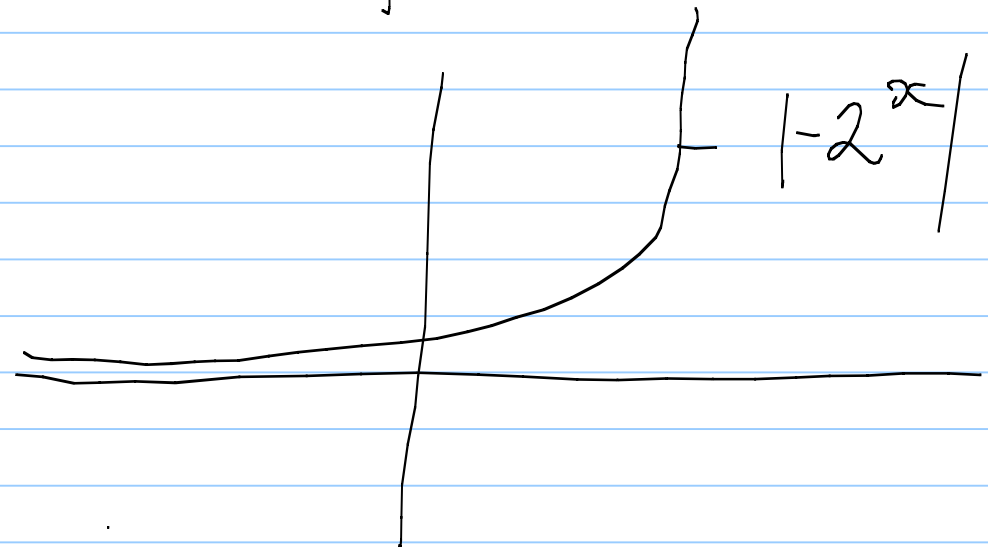
① $f(x) = x - 1$



② $f(x) = x^2 - 1$



③ $f(x) = -2^x$

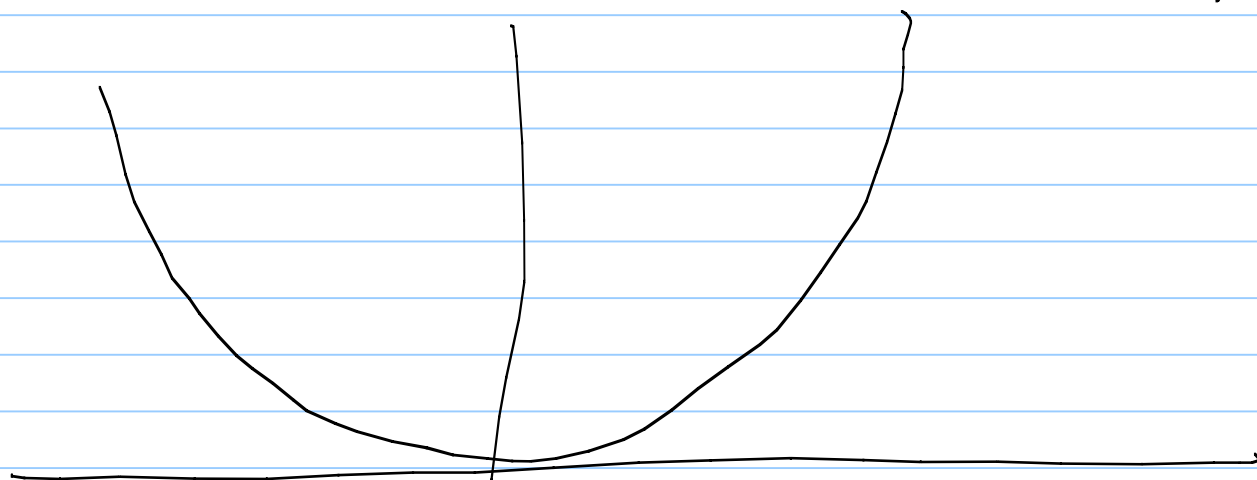


Even and odd functions.

- If a function f satisfies $f(-x) = f(x)$ for every number x in its domain then f is called a even function

Example $f(x) = x^2$

$$f(-x) = (-x)^2 = x^2 = f(x)$$



The graph is symmetric with respect to the y-axis.

- If f satisfies $f(-x) = -f(x)$ for every number x in its domain, then f is called an odd function.

Example $f(x) = x^3$

$$f(-x) = (-x)^3 = -x^3 = -f(x)$$



The graph is symmetric about the origin.

Example Are the following functions even, odd or neither?

① $f(x) = \frac{-x}{x^2 + 1}$

$$f(-x) = \frac{-(-x)}{(-x)^2 + 1}$$

$$= \frac{x}{x^2 + 1}$$

$$= -f(x)$$

∴ odd

$$\textcircled{2} \quad g(t) = -t^2 - 1$$

$$\begin{aligned} g(-t) &= -(-t)^2 - 1 \\ &= -t^2 - 1 \\ &= g(t) \end{aligned}$$

Even.

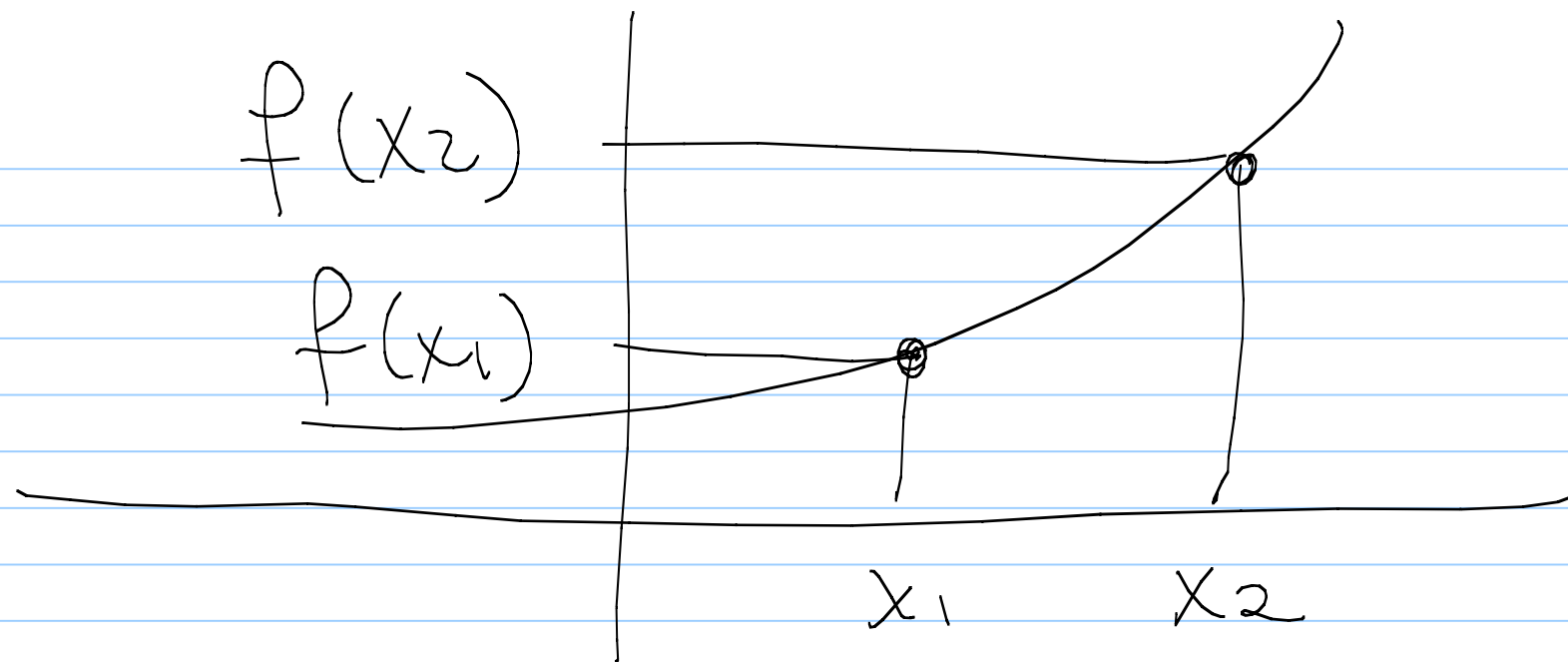
$$\textcircled{3} \quad h(z) = z^2 + 3z + 1$$

$$\begin{aligned} h(-z) &= (-z)^2 + 3(-z) + 1 \\ &= z^2 - 3z + 1 \end{aligned}$$

neither

Increasing and decreasing functions

A function f is called increasing on an interval I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ in I .



A function is called decreasing on I if

$$f(x_1) > f(x_2) \text{ whenever } x_1 < x_2 \text{ in } I$$

